

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------|---|
| 1. Scalar D | (A) The number of independent columns of a matrix. |
| 2. Magnitude F | (B) A vector whose length is exactly 1. |
| 3. Unit vector B | (C) All vectors you can reach as $A\mathbf{x}$ — the span of a matrix's columns. |
| 4. Linear combination G | (D) A single number that scales (stretches or flips) a vector. |
| 5. Span H | (E) $\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2$; it measures length and angle. |
| 6. Dot product E | (F) The length of a vector, $\sqrt{v_1^2 + v_2^2}$. |
| 7. Column space C | (G) A sum of scaled vectors, $c\mathbf{v} + d\mathbf{w}$. |
| 8. Rank A | (H) The set of <i>all</i> linear combinations of a set of vectors. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Two nonzero vectors are **perpendicular** exactly when (A) their sum is **0** (B) $\mathbf{v} \cdot \mathbf{w} = 0$ ← (C) $\mathbf{v} \cdot \mathbf{w} = 1$ (D) they have equal length
- The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ span (A) a single point (B) **a line** ← (C) the whole plane (D) all of space
- The product $A\mathbf{x}$ is (A) **a combination of the columns of A** ← (B) a combination of the rows of A (C) always **0** (D) the length of $\mathbf{x}$
- The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank (A) 0 (B) **1** ← (C) 2 (D) 4
- The unit vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ has length (A) 0 (B) **1** ← (C) $\|\mathbf{v}\|$ (D) $\|\mathbf{v}\|^2$
- A vector $\mathbf{b}$ lies in the column space of A exactly when (A) $\mathbf{b} = \mathbf{0}$ (B) **$A\mathbf{x} = \mathbf{b}$ has a solution** ← (C) $\mathbf{b}$ is a unit vector (D) $\mathbf{b}$ is perpendicular to every column

Teacher Note

Item 2: parallel columns ($\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix}$) span only a line. Item 4: rank 1 because column 2 is $2 \times$ column 1. Watch for students picking (A) “rows” in item 3 — reinforce that $A\mathbf{x}$ combines *columns*.

Part C — Short Answer & Computation (30 pts)

Show your work.

- Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute (a) $\mathbf{v} + \mathbf{w}$, (b) $2\mathbf{v} - \mathbf{w}$, (c) $\|\mathbf{v}\|$. (a) $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$; (b) $\begin{bmatrix} 6 \\ 8 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$; (c) $\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.
- Write the unit vector that points the same direction as $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. $\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$ (divide by the length 5).
- Let $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$, then state whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular and how you know. $\mathbf{u} \cdot \mathbf{v} = (4)(-1) + (2)(2) = -4 + 4 = 0$. **Yes, perpendicular: the dot product is 0.**
- Find weights c and d so that $c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$. $2c + d = 5$, $c + 3d = 5 \Rightarrow c = 2$, $d = 1$. **Check:** $2\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.
- Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Compute $A\mathbf{x}$ as a combination of the columns of A . $A\mathbf{x} = 2\begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$.
- Factor $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$ as $A = CR$. Which columns are independent? Write C , build the recipe R , and give the rank. **Columns 1 and 2 are independent; column 3** $= \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. **So** $C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, **rank 2.**
- Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. For each target, say whether it is reachable as $A\mathbf{x}$ (is it in the column space?) and justify in one line. (a) $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ (b) $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **Both columns are multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so $C(A)$ is that line. (a) Reachable: $\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. (b) Not reachable: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is not a multiple of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (it is off the line).**

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- A matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ has columns $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$. (a) Is the column space a line or the whole plane? How do you know? (b) Is $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ reachable? If so, find the weights. (c) If the second column were changed to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, what would happen to the column space, and why? (a) **The whole plane: the two columns are not multiples of each other (independent), so their span is all of $\mathbb{R}^2$.** (b) **Reachable** — $c\begin{bmatrix} 1 \\ 1 \end{bmatrix} + d\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ gives $c + 2d = 4$, $c + 3d = 5 \Rightarrow d = 1$, $c = 2$. (c) **Then column 2** $= \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **is parallel to column 1, so the columns collapse to one direction and $C(A)$ shrinks to a line (rank drops to 1).**
- Two viewers' taste profiles are $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$ and explain what the result says about how similar the two viewers' tastes are, using what you know about the sign of a dot product. $\mathbf{u} \cdot \mathbf{v} = (2)(-1) + (1)(2) = 0$. **A zero dot product means the profiles are perpendicular — neither aligned nor opposed — so the two viewers' tastes are essentially unrelated (no shared preference direction). A positive dot would mean similar tastes, a negative dot opposite tastes.**

Teacher Note

Scoring Part D (5 pts each). Award full credit only when the student both gives the correct result and justifies it. D1: 2 pts (a) independence $\Rightarrow$ plane; 2 pts (b) correct weights $c = 2, d = 1$; 1 pt (c) parallel $\Rightarrow$ line/rank 1. D2: 2 pts the computation $\mathbf{u} \cdot \mathbf{v} = 0$; 3 pts interpreting $0 \Rightarrow$ perpendicular $\Rightarrow$ unrelated, with the sign meaning. Full-credit language for the arc: *combine columns $\rightarrow$ span $\rightarrow$ line vs. plane $\rightarrow$ reachability.*